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Designing a Prototype Relational Database for Renewal Development's House Relocation Workflow

BCIT GEOGRAPHIC INFORMATION SYSTEMS

BOB ZHANG, A01458419 | ESHA KARIM, A01312891 – SET A

Abstract

Renewal Development offers a sustainable alternative to demolition by relocating houses; however, their current operations are hindered by decentralized data storage across multiple platforms. This project addresses these inefficiencies by designing and prototyping a centralized Relational Database Management System (RDBMS) to streamline the management of spatial, engineering, and administrative records. The primary objective is to consolidate disparate information sources—including GIS layers, CRM records, and documentation—into a unified, normalized structure that supports site selection and route planning.

The project methodology proceeds through five distinct phases: initiation and research, comprehensive database design (including Entity-Relationship modelling), prototype development using SQL and Python integrations, documentation, and project management. Key deliverables will include a comparative RDBMS research report, a functional database prototype with basic integrations for ArcGIS and Zoho CRM, and a complete system architecture guide. The project is scheduled to conclude on May 23, 2026, with a total estimated effort of 720 hours divided between two students. This solution will provide the foundational infrastructure necessary to enhance workflow efficiency and support the sponsor's expanding operations.